AI-scheduled interventions for high-patient receptivity in psycho-oncology



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Contents

1 Introduction **1**

1.1 Background . . . 1

1.1.1 University's description . . . 1

1.1.1.1 Dataset . . . 1

1.1.2 Similar work . . . 1

1.2 Problem definition . . . 2

1.2.1 Problem context . . . 2

1.2.2 Problem justification . . . 2

1.2.3 Research question . . . 2

1.2.4 Hypothesis . . . 2

1.3 Objectives . . . 2

1.3.1 Main objective . . . 2

1.3.2 Specific objectives . . . 2

1.4 Methodology . . . 3

1.5 Constraints . . . 3

1.6 Structure . . . 3

2 Literature review **5**

2.1 Sensing mental health . . . 5

2.1.1 Mobile health . . . 5

2.1.2 Active digital phenotyping . . . 5

2.1.2.1 Ecological Momentary Assessments . . . 6

2.1.2.2 Interventions . . . 6

2.1.3 Passive digital phenotyping . . . 6

2.2 Machine learning in healthcare . . . 6

2.2.1 Sensor inference 6
2.2.2 Ethical considerations 6
2.3 State of the art 7
2.3.1 Predicting receptivity 7
2.3.2 Time series classification 7

3 Research planning matrix **9**

References **11**

1

Introduction

The introduction of this dissertation must include and explain and the key terms related: AI-scheduled interventions for high-patient receptivity in psycho-oncology.

1.1 Background

1.1.1 University's description

Provide essential information about the University of Virginia and its involvement in psychology.

1.1.1.1 Dataset

The data consists of around 400 cancer survivors who were followed for 35 days, during which Ecological Momentary Assessments were taken up to 3 times per day, in addition to sensor data from the subjects' smartphones.

1.1.2 Similar work

Acknowledge recent research conducted over the same dataset also supervised by Dr. Pepijn Van De Ven (this work will be released at the end of summer).

1.2 Problem definition

1.2.1 Problem context

The availability for interventions of cancer survivors will change on a person-to-person basis. Therefore, there is no consensus on what the best time is for AI interventions for high-patient receptivity is.

1.2.2 Problem justification

Research shows that psychological interventions after successful cancer treatment can be sources of stress. Using AI to predict when it's the best time approach people has been explored in other domains, but not in psycho-oncology.

1.2.3 Research question

The main question that this dissertation addresses is "Does delivering interventions at a machine learning (ML)-detected time lead to higher receptivity than delivering interventions at a random time?"

1.2.4 Hypothesis

The expectation is that patients with a good mood will be more receptivity to EMA interventions.

1.3 Objectives

1.3.1 Main objective

The main objective behind this dissertation is to answer the research question proposed in section 1.2.3.

1.3.2 Specific objectives

In order to address the research question, this work focuses on the following issues:

- Infer mood/stress or activities (walking, sitting, driving...) from sensor data.
- Investigate then determine the best time for interventions based on EMAs that will get higher receptivity.

1.4 Methodology

In order to address this question the following approach was taken: Use Deep Learning techniques and EMA data to predict the best time to approach subjects that recently finished their cancer treatment.

1.5 Constraints

Comment of limitations related to the dataset or the scope of the research. Work can only be conducted on authorised computers (hardware limitations). subjects from the dataset were allowed to turn off most of the smartphone sensors.

1.6 Structure

Explain here briefly the chapters that form this dissertation (thesis outline). Chapter 2: literature review, Chapter 3: methodology, Chapter 4: findings, Chapter 5: conclusions, Chapter 6: future work and recommendations.

2

Literature review

Literature review of smart healthcare systems. Concepts related to human psychology, digital phenotyping and machine learning will be explained. Projects in this domain will be evaluated too.

2.1 Sensing mental health

Smartphones and wearable devices created opportunities to better monitor mental health.

2.1.1 Mobile health

Explain important concepts of mobile health (mHealth): engagement, interruptibility, availability and Receptivity.

2.1.2 Active digital phenotyping

Sources of health and behavioural data that require user engagement. Assessments are used for this purpose, however, interventions can in a way be already a way to assess individuals.

2.1.2.1 Ecological Momentary Assessments

Ecological momentary assessments (EMAs) are a well-studied method that has proven useful in understanding an individual's state of mind (Künzler et al. 2019).

2.1.2.2 Interventions

Discuss some frameworks: just-in-time interventions (JITIs), just-in-time adaptive interventions (JITAIs), Ecological Momentary Interventions (EMIs). Acknowledge that EMAs can be interventions too.

2.1.3 Passive digital phenotyping

Describe other common sources of data that are gathered from the natural environment, with no need of user input (Künzler et al. 2019), (Mishra et al. 2021). Describe recently used wearables for this purpose (King et al. 2024).

2.2 Machine learning in healthcare

Big data and Electronic Health Records (EHRs) have allowed the implementation of Machine Learning (ML) solution for a variety of health problems (Rani et al. 2025).

2.2.1 Sensor inference

Acknowledge the smartphone data processing done by pioneers (Sarker et al. 2014) and a recent approach with a continual deep learning predictor (Katevas et al. 2017).

2.2.2 Ethical considerations

Address important considerations such as: data privacy and security, consent and transparency and algorithmic accountability (Rani et al. 2025).

2.3 State of the art

Explain how recent studies in psycho-oncology justify efforts to effectively intervene subjects with care (Heathcote et al. 2022).

2.3.1 Predicting receptivity

Express the differences in the datasets, machine learning models applied, architectures and results (Künzler et al. 2019), (Mishra et al. 2021), (King et al. 2024).

2.3.2 Time series classification

This space is now dominated by deep learning solutions like CNNs and transformers (Wang et al. 2025).

2. LITERATURE REVIEW

3

Research planning matrix

3. RESEARCH PLANNING MATRIX

Table 3.1: Research planning matrix

Research Topic	AI scheduled interventions for high-patient receptivity in psycho-oncology.
Identified Research Gap and Justification	Research shows that psychological interventions after successful cancer treatment can be sources of stress. ~~Using AI to predict when it's the best time approach people has been explored in other domains, but not in psycho-oncology.~~ Only recently have efforts to use AI for availability prediction in the domain of psycho-oncology started, this dissertation seeks to expands this research space.
Research Problem	~~There's no consensus on what's the best time for interventions for high-patient receptivity~~ A body of work started last year with three students of the MSc in AI to investigate what's the best time for interventions for high-patient receptivity over the same EARS dataset (Kukreja 2026), (Malgond 2026), (Sharma 2026). Due to time constraints, the explored research space was limited and there are opportunities to continue their work.
Research Goals	~~* Infer mood/stress or activities (walking, sitting, driving...) from sensor data~~ ~~* Investigate and then determine the best time for interventions based on EMAs that will get higher receptivity~~ * Incorporate GPS depersonalised features to the input space for the model(s) * Modify preprocessing pipeline to utilise the questionnaires answered every three hours in the dataset, as oppose to only the weekly questionnaires. * Considering the previous objective, explore different window sizes, beyond the 48 hour window used by the three researchers who worked on the dataset.
Research Questions	* Does delivering interventions at a machine learning (ML)-detected time lead to higher receptivity than delivering interventions at a random time? COMMENT: here I'm tempted to use the results of the previous students instead of 'at a random time' but if I change the size of the window, I suppose I can't draw direct comparisons
Research Methods for Achieving This	Use Deep Learning techniques and EMA data to predict the best time to approach subjects that recently finished their cancer treatment.
Type of Data Collected, Data Analysis Methods, and Triangulation Potential	The data consists of around 400 cancer survivors who were followed for 35 days, during which Ecological Momentary Assessments were taken up to 3 times per day and also on a weekly basis, in addition to sensor data from the subjects' smartphones.
Possible Expected Results and Hypotheses	~~The expectation is that patients with a good mood will display higher receptivity to EMA interventions~~ The addition of GPS features and a smaller window size over more frequent labels should improve the learning.

References

Heathcote, L. C., Cunningham, S. J., Webster, S. N., Tanna, V., Mattke, E., Loecher, N., Spunt, S. L., Simon, P., Dahl, G., Walentynowicz, M., Murnane, E., Tutelman, P. R., Schapira, L., Simons, L. E. and Mueller, C. (2022), 'Smartphone-based ecological momentary assessment to study "scanxiety" among adolescent and young adult survivors of childhood cancer: A feasibility study', Psycho-oncology (Chichester, England) **31**(8), 1322–1330. 7

Katevas, K., Leontiadis, I., Pielot, M. and Serrà, J. (2017), 'Practical processing of mobile sensor data for continual deep learning predictions', arXiv.org . 6

King, Z. D., Yu, H., Vaessen, T., Myin-Germeys, I. and Sano, A. (2024), 'Investigating receptivity and affect using machine learning: Ecological momentary assessment and wearable sensing study', JMIR mHealth and uHealth **12**, e46347–. 6, 7

Kukreja, S. (2026), Predicting intervention availability of a user based on ecological momentary assessment and smartphone sensor data using convolutional neural networks, MSc dissertation, University of Limerick, Limerick, Ireland. Unpublished. MSc in Artificial Intelligence and Machine Learning, Department of Electronic and Computer Engineering. Supervised by Prof. Pepijn Van de Ven. 10

Künzler, F., Mishra, V., Kramer, J.-N., Kotz, D., Fleisch, E. and Kowatsch, T. (2019), 'Exploring the state-of-receptivity for mhealth interventions', Proceedings of ACM on interactive, mobile, wearable and ubiquitous technologies **3**(4), 1–27. 6, 7

Malgond, A. (2026), CNN-based behavioural availability prediction from smartphone sensor data, MEng final project report, University of Limerick, Limerick, Ireland. Unpublished. LK489 MEng in Computer Vision and Artificial Intelligence, Department of Electronic and Computer Engineering. Supervised by Prof. Pepijn Van de Ven. 10

REFERENCES

Mishra, V., Künzler, F., Kramer, J.-N., Fleisch, E., Kowatsch, T. and Kotz, D. (2021), 'Detecting receptivity for mhealth interventions in the natural environment', Proceedings of ACM on interactive, mobile, wearable and ubiquitous technologies **5**(2), 1–24. 6, 7

Rani, S., Kumar, R., Panda, B. S., Kumar, R., Muften, N. F., Abass, M. A. and Lozanović, J. (2025), 'Machine learning-powered smart healthcare systems in the era of big data: Applications, diagnostic insights, challenges, and ethical implications', Diagnostics (Basel) **15**(15), 1914–. 6

Sarker, H., Sharmin, M., Ali, A. A., Rahman, M. M., Bari, R., Hossain, S. M. and Kumar, S. (2014), Assessing the availability of users to engage in just-in-time intervention in the natural environment, in 'Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing', ACM, New York, NY, USA, pp. 909–920. 6

Sharma, P. (2026), Transformer-based modelling of smartphone behavioural sequences for intervention availability prediction, MEng final project report, University of Limerick, Limerick, Ireland. Unpublished. LK489 MEng in Computer Vision and Artificial Intelligence, Department of Electronic and Computer Engineering. Supervised by Prof. Pepijn Van de Ven. 10

Wang, Y., Wu, H., Dong, J., Liu, Y., Wang, C., Long, M. and Wang, J. (2025), 'Deep time series models: A comprehensive survey and benchmark'. 7